HIMANSHU SINGH

Backend & AI Engineer | Spring Boot | Distributed Systems | FastAPI | Docker

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Portfolio | GitHub: Alieno1 | LinkedIn: Himanshu Singh

EDUCATION

•National Institute of Technology, Surat *Graduation: May 2026*

Bachelor of Technology in Computer Science and Engineering Surat, India

EXPERIENCE

- **Software Development Engineer Intern** *Jan 2026 – June 2026*

  Jainam Broking Limited Surat, India

  – **Market Data & Strategy:** Engineered an end-to-end pipeline via **Apache Kafka** to ingest **120k+ tick messages/sec**; implemented **sliding-window aggregates** for real-time OHLC generation and a **custom Java consumer implementation** for **TimescaleDB Hypertables** sync, achieving **sub-50ms p99 latency**.

  – **Quantitative Analytics:** Integrated a high-performance simulation engine to backtest **SMA/Momentum strategies** against historical data; automated risk-reward metric generation and equity curve reporting by **directly updating database table views**.

  – **Order Management (OMEX):** Architected an ingestion layer mimicking institutional trading platforms (**NEST/ODIN**); validated system resilience via stress-testing **100,000+ messages** with **zero loss** and optimized consumer-group rebalancing.

  – **Risk Management Engine:** Designed a fault-tolerant calculator for **MCX margin requirements** using **SPAN files**; leveraged **CockroachDB** for distributed reliability and **Redis distributed cache** for sub-millisecond margin lookups.

  – **Performance Tuning:** Orchestrated **Quarkus microservices** using **Reactive Streams (Mutiny)** to reduce memory footprint by **40%**; fine-tuned **Kafka session.timeout/heartbeat** intervals to ensure **zero-lag streaming** during peak volatility.

TECHNICAL SKILLS

- **Backend & AI:** Spring Boot, FastAPI, Quarkus, Mutiny, Microservices, LangChain, Gemini API, RAG, ChromaDB
- **Databases:** MySQL, Redis, SQL Optimization, Indexing, Transactions, TimescaleDB, CockroachDB
- **Infra & Testing:** Docker, Nginx, CI/CD (GitHub Actions), Linux, Pytest, Pytest-Mock
- **Languages:** Java, Python, C/C++, JavaScript
- **Tools & Concepts:** Git, Postman, Distributed Systems, DSA, OS, CN, Cloud Computing, System Design

PROJECTS

- **Enterprise AI Operations Copilot** *Sep 2026*

  Python, LangChain, Google Gemini, ChromaDB, Streamlit, Pytest GitHub Repo | Live Demo

  – Engineered a decoupled **Agentic AI Copilot** using **LangChain** and **Google Gemini** to autonomously route user queries to distinct operational backend tools.

  – Architected a **Retrieval-Augmented Generation (RAG)** pipeline with **ChromaDB** and **Hugging Face** embeddings, enabling zero-hallucination semantic search across unstructured enterprise policies.

  – Validated system resilience via a comprehensive **Pytest** suite, heavily mocking LLM API responses and VectorStores to achieve complete test coverage without quota exhaustion.

- **Auto-Director: Agentic Micro-Drama Engine** *Jul 2026*

  Python, Multi-Agent System, MoviePy, FFmpeg, Gemini, ElevenLabs GitHub Repo | Live Demo

  – Architected an autonomous **5-agent pipeline** (Scriptwriter, Voice, Visual, Subtitle, Director) to convert raw story text into 9:16 vertical micro-dramas with audio-visual synchronization.

  – Engineered **multi-tier provider fallbacks** (ElevenLabs $\rightarrow$ gTTS $\rightarrow$ espeak; Gemini $\rightarrow$ Pollinations $\rightarrow$ Stock Photos) with intelligent HTTP 4xx status filters to bypass API rate limits.

  – Developed dynamic font resolution for Hindi Devanagari/English scripts and optimized PIL stroke rendering by **49**$\times$ for zero-tofu subtitle overlays.

- **Hotel Automation Controller** *May 2026*

  Java, Spring Boot, System Architecture GitHub Repo

  – Developed a **Java 17/Spring Boot** backend using **SOLID principles** to accurately map external sensor motion signals to specific device states.

  – Engineered a concurrent **power-budgeting algorithm** via **ScheduledExecutorService** to strictly enforce per-floor load constraints.

  – Built **RESTful APIs** for scale and achieved high reliability through comprehensive **JUnit/Mockito** testing.

KEY COURSES

- Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Distributed Systems, Cloud Computing, Information Security, High Performance Computing, Machine Learning, Deep Learning